Veer Project Plan

Course: Computer Science Senior Design I

Advisor: Dr. Jillian Aurisano

Team: Grady Algire, Brady Cooper, Briar Elliot, Dominic Rowland, Aiden Ward

Assignments #5 and #6 | October 28, 2026

Project goal: Convert driver speech into routing constraints for hands-free navigation

Names and IDs match D0 to D2. T = task, M = milestone, B = protected buffer block, RK = risk.

1. Task List (Assignment #5)

T1. Develop the C1 Mobile Navigation app shell: destination entry, map, and turn-by-turn guidance (I1b, I1c). Owner: Grady Algire.

T2. Develop the C1 Mobile Navigation trip state machine and its I2/I4 calls. Owner: Grady Algire.

T3. Develop the Configuration component and GET /config/client. Owner: Grady Algire.

T4. Develop the backend on Render: Express, PostgreSQL, and CI. Owner: Grady Algire.

T5. Test the full Veer system end to end on supervised test drives (US-01 to US-05). Owner: Grady Algire.

T6. Obtain written proof of the E2 AI Provider no-retention setting and pick the audio format (G-03, G-08). Owner: Briar Elliot.

T7. Develop the C2 Speech Understanding I2 endpoint with E2 transcription. Owner: Briar Elliot.

T8. Develop C2 Speech Understanding constraint extraction and allow-list check. Owner: Briar Elliot.

T9. Develop C2 Speech Understanding "okay" / "no" replies and short confirmations. Owner: Briar Elliot.

T10. Validate C2 Speech Understanding accuracy with accent samples (US-05). Owner: Briar Elliot.

T11. Research E3 Google Maps Platform terms for in-app guidance and get API keys (G-04). Owner: Brady Cooper.

T12. Develop the C3 Route Planning I4 endpoint and E3 routes client. Owner: Brady Cooper.

T13. Develop C3 Route Planning food and business stop insertion (US-01). Owner: Brady Cooper.

T14. Develop C3 Route Planning avoid highways and tolls, and the next route on "no". Owner: Brady Cooper.

T15. Test C3 Route Planning edge cases EC-09 to EC-12. Owner: Brady Cooper.

T16. Investigate E4 Weather Provider data and propose rain settings (G-02). Owner: Dominic Rowland.

T17. Develop the E4 Weather Provider client and rain avoidance in C3 (US-02). Owner: Dominic Rowland.

T18. Investigate on-device "Veer, stop listening" and a C1 Mobile Navigation foreground service (G-01, G-06). Owner: Dominic Rowland.

T19. Develop C1 Mobile Navigation mic control, listening indicator, and one-tap mute (I1a, US-04). Owner: Dominic Rowland.

T20. Develop the Notifications component: trip notice, driver notices, and budget alerts. Owner: Dominic Rowland.

T21. Develop the C4 Usage Monitor database tables and retention job (G-05). Owner: Aiden Ward.

T22. Develop C4 Usage Monitor checkQuota and recordUsage (I7 to I9). Owner: Aiden Ward.

T23. Develop the Authentication component: register, refresh, and Keystore storage. Owner: Aiden Ward.

T24. Develop the Logging component with redaction. Owner: Aiden Ward.

T25. Develop the C4 Usage Monitor usage report and spending caps (US-03, G-07). Owner: Aiden Ward.

2. Milestones List (Assignment #6)

ID	Milestone	Date	Shows	How verified	By whom	Status
M-01	Team roster and signed team contract	Sep 3	Team formed; roles agreed.	Submitted on Canvas.	Instructor	Complete
M-02	Topic statement, bios, and reflective essay	Sep 18	Project idea; member skills.	Files in repo; graded.	Advisor, instructor	Complete
M-03	Advisor named and constraints essay	Sep 18	Advisor; design constraints.	File in repo; graded.	Advisor, instructor	Complete
M-04	User stories and use cases	Sep 26	US-01 to US-05.	File in repo; graded.	Advisor, instructor	Complete
M-05	Design D0 high-level design	Oct 2	Components, interfaces, owners.	Files in repo; graded.	Advisor, instructor	Complete
M-06	Design D1 detailed design part 1	Oct 8	C2 and C3 design, tech stack.	Files in repo; graded.	Advisor, instructor	Complete
M-07	Design D2 detailed design part 2 and peer review	Oct 14	C1, C4, glue design; edge cases.	Files in repo; peer review.	Peer group, advisor	Complete
M-08	Task timeline and effort matrix (this document)	Oct 28	Tasks, dates, effort, risks.	Tasklist.md and Planning/ in repo.	Advisor, instructor	Submitted
M-09	Requirements and technical specification documents	Nov 4	Requirements traced to design.	Documents submitted.	Advisor, instructor	Planned
M-10	Test plan draft	Nov 11	How each requirement is tested.	Draft submitted.	Advisor, instructor	Planned
M-11	Ethics and responsible-use statement	Nov 18	Privacy, licensing, AI use.	Statement submitted.	Advisor, instructor	Planned
M-12	Research gaps closed (G-01 to G-04, G-06, G-08)	Nov 24	Open design questions answered.	Findings in repo; D2 gaps closed.	Advisor; Team	Planned
M-13	Presentation draft and advisor feedback	Nov 25	Slides and demo plan.	Advisor reviews draft.	Advisor	Planned
M-14	Team oral presentation	Dec 2	Topic, design, and plan.	Presented in class.	Instructor, class	Planned
M-15	Backend live on Render with C4 tables	Dec 2	Backend and C4 tables live.	Health check passes; CI green.	Team	Planned
M-16	Design report, design notebook, team and self assessment	Dec 9	Fall progress against this plan.	Report, notebook, assessments.	Advisor, instructor	Planned
M-17	Spring platform ready: auth and logging	Feb 1	Login and redacted logs work.	Scripted smoke test passes.	Team	Planned
M-18	Speech-to-constraint demo	Feb 12	Speech becomes a valid constraint.	8 of 10 test phrases correct.	Team; advisor	Planned
M-19	Route proposal demo (US-01)	Feb 15	Phone gets a route with a food stop.	Live demo on a phone.	Team; advisor	Planned
M-20	Alpha: full voice loop in a parked car	Mar 3	Full voice loop; mute under 1 s.	Timed demo in a parked car; video.	Team; advisor	Planned
M-21	Beta: accuracy and edge cases checked	Mar 15	Accent and edge-case tests pass.	Test reports in repo.	Team; advisor	Planned
M-22	Usage report and caps enforced (US-03)	Mar 16	Usage report and budget alerts work.	Test budget hits 80 and 100 percent.	Team	Planned
M-23	Feature freeze after test drives	Apr 2	Every scope item works on real drives.	Drive log, each story pass/fail.	Team; advisor	Planned
M-24	Final demo build tagged	Apr 8	Expo build ready.	v1.0 tag installs on a clean phone.	Team; advisor	Planned
M-25	Senior Design Expo demo	Apr 15	Live demo for visitors and judges.	Expo judging.	Judges, advisor	Planned

3. Table 1: Timeline (Assignment #6)

Buffer is 20 percent per task, from Work done to Completion. Yellow rows (B) are protected blocks before major milestones. CP = critical path. No work is planned over Thanksgiving, finals, winter break, or spring break.

ID	Task or milestone	Owner	Start	Work done	Buffer	Completion	Depends on	CP / Risk
T4	Develop the backend on Render: Express, PostgreSQL, and CI	Grady	Nov 2	Nov 10	2 d	Nov 12	None	RK-05
T6	Obtain written proof of the E2 AI Provider no-retention setting and pick the audio format (G-03, G-08)	Briar	Nov 2	Nov 9	2 d	Nov 11	None	RK-01
T11	Research E3 Google Maps Platform terms for in-app guidance and get API keys (G-04)	Brady	Nov 2	Nov 9	2 d	Nov 11	None	RK-02
T18	Investigate on-device "Veer, stop listening" and a C1 Mobile Navigation foreground service (G-01, G-06)	Dominic	Nov 2	Nov 9	2 d	Nov 11	None	RK-03
M-09	**Milestone: Requirements and technical specification documents**					Nov 4		
M-10	**Milestone: Test plan draft**					Nov 11		
T16	Investigate E4 Weather Provider data and propose rain settings (G-02)	Dominic	Nov 12	Nov 19	2 d	Nov 23	None	RK-09
T3	Develop the Configuration component and GET /config/client	Grady	Nov 13	Nov 23	2 d	Nov 30	T4	
T21	Develop the C4 Usage Monitor database tables and retention job (G-05)	Aiden	Nov 13	Nov 24	2 d	Dec 1	T4	
M-11	**Milestone: Ethics and responsible-use statement**					Nov 18		
M-12	**Milestone: Research gaps closed (G-01 to G-04, G-06, G-08)**					Nov 24	T18, T6, T11, T16	
M-13	**Milestone: Presentation draft and advisor feedback**					Nov 25		
B-01	Protected block before M-16	Team	Dec 2	Dec 8	5 d	Dec 8		
M-14	**Milestone: Team oral presentation**					Dec 2		
M-15	**Milestone: Backend live on Render with C4 tables**					Dec 2	T4, T21	
M-16	**Milestone: Design report, design notebook, team and self assessment**					Dec 9	T21, T3, T18	
T1	Develop the C1 Mobile Navigation app shell: destination entry, map, and turn-by-turn guidance (I1b, I1c)	Grady	Jan 11	Jan 22	2 d	Jan 26	T18, T11	**CP RK-02**
T7	Develop the C2 Speech Understanding I2 endpoint with E2 transcription	Briar	Jan 11	Jan 22	2 d	Jan 26	T6, T4	RK-01, RK-04
T12	Develop the C3 Route Planning I4 endpoint and E3 routes client	Brady	Jan 11	Jan 25	2 d	Jan 27	T11, T4	RK-02
T17	Develop the E4 Weather Provider client and rain avoidance in C3 (US-02)	Dominic	Jan 11	Jan 25	2 d	Jan 27	T16	RK-09
T23	Develop the Authentication component: register, refresh, and Keystore storage	Aiden	Jan 11	Jan 20	2 d	Jan 22	T4	
T24	Develop the Logging component with redaction	Aiden	Jan 25	Jan 28	1 d	Jan 29	T3	
T2	Develop the C1 Mobile Navigation trip state machine and its I2/I4 calls	Grady	Jan 27	Feb 9	2 d	Feb 11	T1, T23	**CP RK-08**
T8	Develop C2 Speech Understanding constraint extraction and allow-list check	Briar	Jan 27	Feb 9	2 d	Feb 11	T7	RK-04
T13	Develop C3 Route Planning food and business stop insertion (US-01)	Brady	Jan 28	Feb 10	2 d	Feb 12	T12	
T22	Develop C4 Usage Monitor checkQuota and recordUsage (I7 to I9)	Aiden	Feb 1	Feb 11	2 d	Feb 15	T21, T3	RK-05
M-17	**Milestone: Spring platform ready: auth and logging**					Feb 1	T24, T23	
T9	Develop C2 Speech Understanding "okay" / "no" replies and short confirmations	Briar	Feb 12	Feb 19	2 d	Feb 23	T8	
T19	Develop C1 Mobile Navigation mic control, listening indicator, and one-tap mute (I1a, US-04)	Dominic	Feb 12	Feb 22	2 d	Feb 24	T2	**CP RK-03**
M-18	**Milestone: Speech-to-constraint demo**					Feb 12	T7, T8	

ID	Task or milestone	Owner	Start	Work done	Buffer	Completion	Depends on	CP / Risk
T14	Develop C3 Route Planning avoid highways and tolls, and the next route on "no"	Brady	Feb 15	Feb 23	2 d	Feb 25	T13	
M-19	**Milestone: Route proposal demo (US-01)**					Feb 15	T13, T2	
T25	Develop the C4 Usage Monitor usage report and spending caps (US-03, G-07)	Aiden	Feb 16	Feb 23	2 d	Feb 25	T22, T23	RK-05
T10	Validate C2 Speech Understanding accuracy with accent samples (US-05)	Briar	Feb 24	Mar 3	2 d	Mar 5	T9	RK-04
T20	Develop the Notifications component: trip notice, driver notices, and budget alerts	Dominic	Feb 25	Mar 4	2 d	Mar 15	T22, T1	**CP RK-03**
B-02	Protected block before M-20	Team	Feb 26	Mar 2	3 d	Mar 2		
T15	Test C3 Route Planning edge cases EC-09 to EC-12	Brady	Feb 26	Mar 4	1 d	Mar 5	T14, T17	
M-20	**Milestone: Alpha: full voice loop in a parked car**					Mar 3	T19, T9, T14, T17, T22	
M-21	**Milestone: Beta: accuracy and edge cases checked**					Mar 15	T10, T15, T24	
T5	Test the full Veer system end to end on supervised test drives (US-01 to US-05)	Team	Mar 16	Mar 25	2 d	Mar 29	T19, T9, T14, T17, T24, T20, T25, M-20	**CP RK-08, RK-09**
M-22	**Milestone: Usage report and caps enforced (US-03)**					Mar 16	T20, T25	
B-03	Protected block before M-23	Team	Mar 30	Apr 1	3 d	Apr 1		
M-23	**Milestone: Feature freeze after test drives**					Apr 2	T5	
B-04	Protected block before M-24	Team	Apr 5	Apr 7	3 d	Apr 7		
M-24	**Milestone: Final demo build tagged**					Apr 8	T5, T10, T15	
B-05	Protected block before M-25	Team	Apr 9	Apr 14	4 d	Apr 14		
M-25	**Milestone: Senior Design Expo demo**					Apr 15	T5	

Table 1. Fall dates are 2026; spring dates are 2027.

Critical path. T1 → T2 → T19 → T20 → T5. A slip on any of these moves M-23.

Risks. T6, T11, and T18 come first because they answer RK-01 to RK-03. M-19 and M-20 come before spring break for RK-08. Fall tasks are small because a course deliverable is due every week.

4. Table 2: Effort Matrix (Assignment #6)

Expected hours, (O + 4M + P) / 6. * = primary owner, who has the largest share.

ID	Task	Grady	Brady	Briar	Dominic	Aiden	Total
T1	C1 app shell and guidance	**21.0***	3.7				24.7
T2	C1 trip state machine	**23.2***		2.9	2.9		29.0
T3	Configuration	**12.9***				1.4	14.3
T4	Backend deployment on Render	**12.8***	2.5			1.7	17.0
T5	End-to-end test drives	**18.8***	3.1	3.1	3.1	3.1	31.3
T6	E2 no-retention + audio format			**10.7***			10.7
T7	C2 I2 endpoint + transcription			**22.5***		2.5	25.0
T8	C2 constraint extraction			**24.3***		2.7	27.0
T9	C2 approval replies	1.7		**15.0***			16.7
T10	C2 accent validation			**13.4***		3.3	16.7
T11	E3 Maps terms + keys		**10.3***				10.3
T12	C3 I4 endpoint + Maps client		**24.3***		2.7		27.0
T13	C3 waypoint insertion		**24.3***		2.7		27.0
T14	C3 preferences + next candidate	1.9	**16.8***				18.7
T15	C3 edge-case tests		**12.2***		2.1		14.3
T16	E4 weather thresholds				**10.7***		10.7
T17	E4 client + rain scoring		4.0		**22.9***		27.0
T18	C1 local mute + foreground service	1.3			**11.4***		12.7
T19	C1 mic, mute, MuteError	2.9			**16.1***		19.0
T20	Notifications	1.7			**15.0***		16.7
T21	C4 schema + retention job					**14.3***	14.3
T22	C4 checkQuota + recordUsage		2.3			**20.4***	22.7
T23	Authentication	2.9				**16.1***	19.0
T24	Logging		1.0			**9.3***	10.3
T25	C4 report + caps					**14.3***	14.3
	Total	**101.1**	**104.5**	**91.9**	**89.6**	**89.1**	**476.2**
	Share	**21%**	**22%**	**19%**	**19%**	**19%**	**100%**

Table 2. Effort matrix (hours).

Imbalance. Average is 95 hours. Brady is 10 percent over because he backs up most backend work. Dominic and Aiden are about 6 percent under. Briar is under on purpose (full-time job, RK-06).

Plan. Aiden takes Brady's support on T4 and Dominic takes Brady's support on T22 (5 hours). Then Brady is at 100 and everyone but Briar is within 6 percent. We log hours weekly and recheck at M-17 and M-20.

5. Work Breakdown Structure

WBS ID	Phase or group	Tasks and milestones
1.0	Design (complete)	M-01; M-02; M-03; M-04; M-05; M-06; M-07; M-08
2.0	Research and Setup (Fall 2026)	T3 (Grady); T4 (Grady); T6 (Briar); T11 (Brady); T16 (Dominic); T18 (Dominic); T21 (Aiden)
3.0	Build (Spring 2027)	See 3.1 to 3.5
3.1	C1 Mobile Navigation	T1 (Grady); T2 (Grady); T19 (Dominic)
3.2	C2 Speech Understanding	T7 (Briar); T8 (Briar); T9 (Briar)
3.3	C3 Route Planning (with E4)	T12 (Brady); T13 (Brady); T14 (Brady); T17 (Dominic)
3.4	C4 Usage Monitor	T22 (Aiden); T25 (Aiden)
3.5	Glue components	T20 (Dominic); T23 (Aiden); T24 (Aiden)
4.0	Test and Validation	T10 (Briar); T15 (Brady); T5 (Team)
5.0	Course deliverables and delivery	M-09; M-10; M-11; M-13; M-14; M-16; M-24; M-25

Diagram version: Figure2_WBS.drawio in the Planning folder.

6. Risk Register

ID	Type	Risk	Likely	Impact	Strategy	Mitigation	Owner	Trigger	In Table 1
RK-01	External	OpenAI no-retention setting not confirmed (G-03).	Medium	High	Reduce	Do T6 first. Fallback: transcribe on the phone, send text only.	Briar	No answer by end of T6.	T6, T7
RK-02	External	Maps terms block in-app guidance (G-04).	Medium	High	Transfer	Do T11 before T1. Fallback: hand off to the Google Maps app.	Brady	T11 finds a blocking term.	T11, T1, T12
RK-03	Technical	Android pauses the mic, or voice mute is over 1 s (G-01, G-06).	Medium	High	Reduce	Prototype in T18. Foreground service; tap-only mute fallback.	Dominic	T18 mute over 1 s.	T18, T19, T20
RK-04	Technical	AI is slow or wrong, mainly with accents.	Medium	Medium	Reduce	Approval always required. Test in T10; tune prompts.	Briar	Under 8 of 10 at M-18.	T7, T8, T10
RK-05	External	API costs pass free credits.	Medium	Medium	Reduce	C4 caps before test drives; mocks in tests.	Aiden	Any budget hits 80 percent.	T4, T22, T25
RK-06	Team	Jobs limit hours (Briar full-time, Grady part-time).	High	Medium	Accept	Briar planned lighter; Aiden backs up C2 reviews.	Briar	A task slips over 3 days past buffer.	Table 2
RK-07	Team	Weekly course deliverables, exams, and breaks cut time.	High	Medium	Avoid	Fall tasks kept small; no work in breaks; protected blocks.	Grady	Task open on last day before a break.	B-01 to B-05
RK-08	Technical	C1, C2, C3 do not fit together.	Medium	High	Reduce	Build to D1 contracts with mocks; M-19 and M-20 before break.	Grady	M-19 or M-20 over 3 days late.	T2, T5, M-19, M-20
RK-09	External	No rain on test days (US-02).	High	Low	Accept	Replay recorded rain data.	Dominic	No rain day by start of T5.	T5, T16, T17